

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

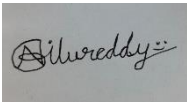
Total Marks: 25

Code of Conduct

I G. Venkata Niranjana Reddy (192525199) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Company Experiences Unpredictable Traffic Spikes During Seasonal Sales

Cloud computing provides **Auto-Scaling** and **Load Balancing** to handle sudden increases in website traffic efficiently. During seasonal events such as Black Friday, Diwali sales, or New Year offers, thousands or even millions of users may access an e-commerce website simultaneously. Without cloud services, servers may become overloaded, causing slow response times or complete website failure.

Auto-Scaling

Auto-scaling automatically adjusts the number of computing resources according to the current workload.

- During high traffic, additional virtual servers are launched automatically.
- When traffic decreases, unnecessary servers are removed.
- This ensures that users always receive good performance.
- Organizations pay only for the resources they use, reducing operational costs.

Load Balancing

Load balancing distributes incoming user requests evenly among multiple servers.

- Prevents any single server from becoming overloaded.
- Improves application availability.
- Reduces response time and increases reliability.
- Supports continuous service even during hardware failures.

Benefits of Auto-Scaling and Load Balancing

- Prevents website crashes during peak traffic.
- Improves application performance.
- Reduces downtime.
- Ensures high availability.
- Lowers operational costs.

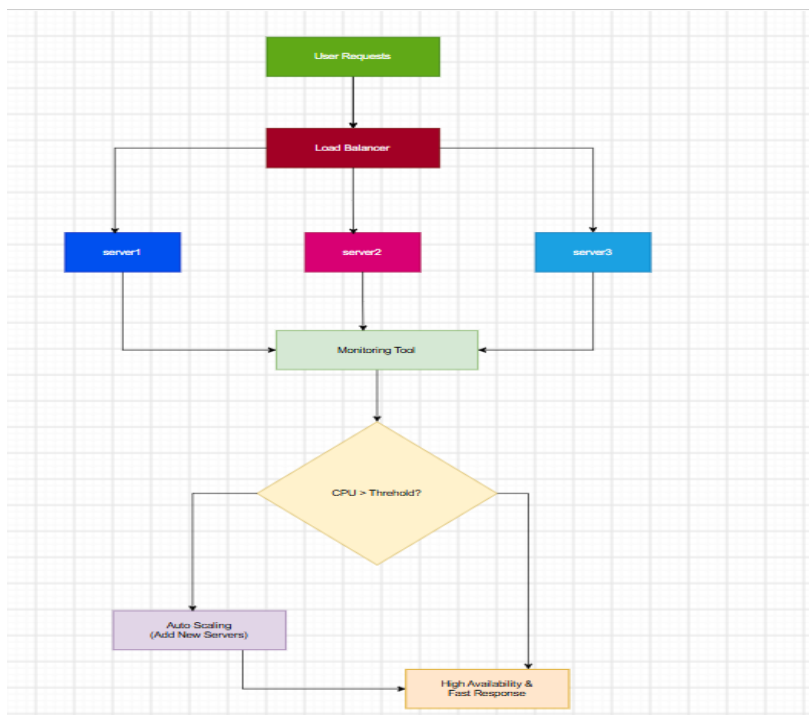
Role of Monitoring Tools

Cloud monitoring tools continuously observe:

- CPU utilization
- Memory usage
- Network traffic

Example

During **Amazon Prime Day**, millions of customers browse products simultaneously. Amazon Web Services (AWS) automatically launches additional servers using Auto Scaling Groups. Elastic Load Balancers distribute customer requests among available servers, while monitoring services such as CloudWatch continuously track system performance. As a result, customers experience fast page loading, uninterrupted shopping, and successful transactions even during massive traffic spikes.



2. Cloud Storage Solutions for Secure Backup Data

Organizations generate huge amounts of business data, customer records, financial information, and application backups. Cloud storage provides a secure, reliable, and scalable solution for storing these backups without maintaining expensive physical storage systems.

Durability

Durability refers to the ability of cloud storage to preserve data for a long period without loss.

Cloud providers ensure durability by:

- Storing multiple copies of data.
- Using redundant storage devices.
- Continuously checking data integrity.
- Automatically replacing failed storage hardware.
- Replicating data across multiple data centers.

This provides durability levels of over **99.999999999%** offered by major cloud providers.

Data Redundancy

Data redundancy means keeping several copies of the same data.

Benefits include:

- Protection against disk failures.
- Recovery from accidental deletion.
- Disaster protection.
- High availability.
- Faster restoration of lost data.

Data may be stored across different availability zones or geographic regions.

Encryption

Encryption protects backup data from unauthorized access.

Data at Rest

Files stored in cloud storage are encrypted using algorithms such as AES-256.

Data in Transit

Data transferred between users and cloud servers is protected using SSL/TLS encryption.

Even if attackers intercept encrypted data, they cannot read it without the encryption key.

Access Control

Cloud providers implement strong security through:

- Identity and Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- User permissions
- Audit logs

Only authorized personnel can access backup files.

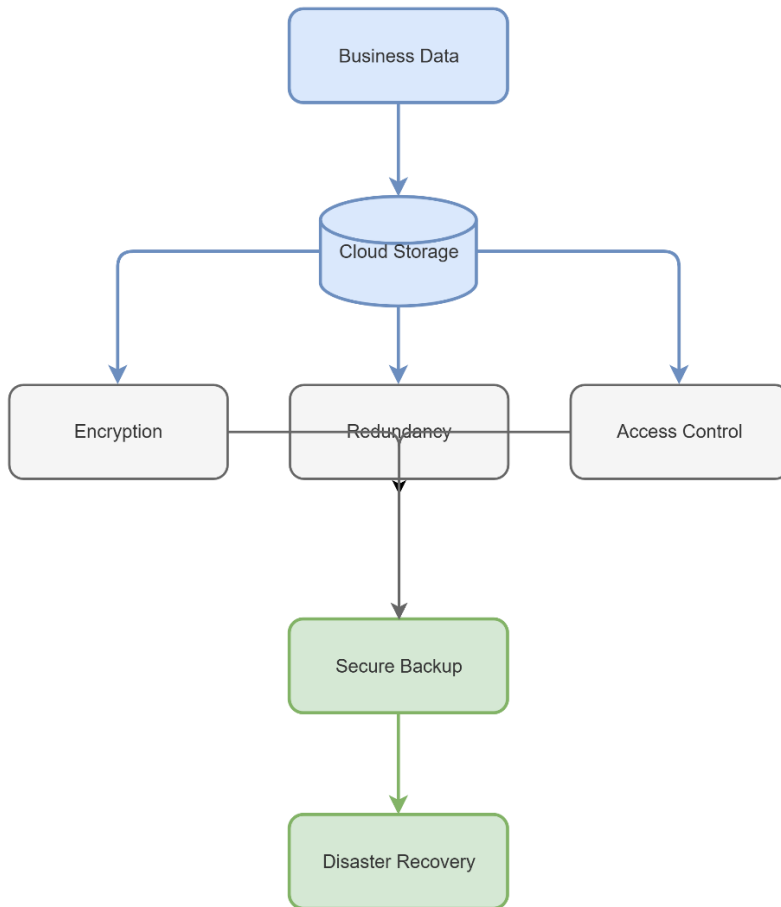
Cost Advantages

Compared with traditional storage systems, cloud storage offers:

- No investment in expensive hardware.
- No maintenance costs.
- No electricity and cooling expenses.
- Pay-as-you-use pricing.
- Easy scalability.
- Reduced administrative workload.

Disaster Recovery Example

Suppose a company's headquarters is destroyed by fire. Since all databases and application backups are stored in cloud storage, the IT team can restore systems from cloud backups within a short period. Employees can continue operations from another location with minimal business interruption.



3. Global Application Deployment Using CDN

A mobile application used worldwide must provide fast access regardless of the user's location. Cloud providers use **Content Delivery Networks (CDNs)** to improve application performance and reduce latency.

A Content Delivery Network is a collection of geographically distributed servers that cache application content close to users.

Instead of fetching data from the original server every time, users receive content from the nearest CDN server.

Benefits of CDN

- Faster page loading.
- Lower latency.
- Reduced bandwidth consumption.

- Better application performance.
- Improved user experience.

Edge Servers

Edge servers are CDN servers located near users.

Their functions include:

- Storing frequently accessed files.
- Handling regional traffic.
- Minimizing delays.

Since requests travel shorter distances, applications become much faster.

Global Availability

Cloud providers maintain data centers across multiple regions.

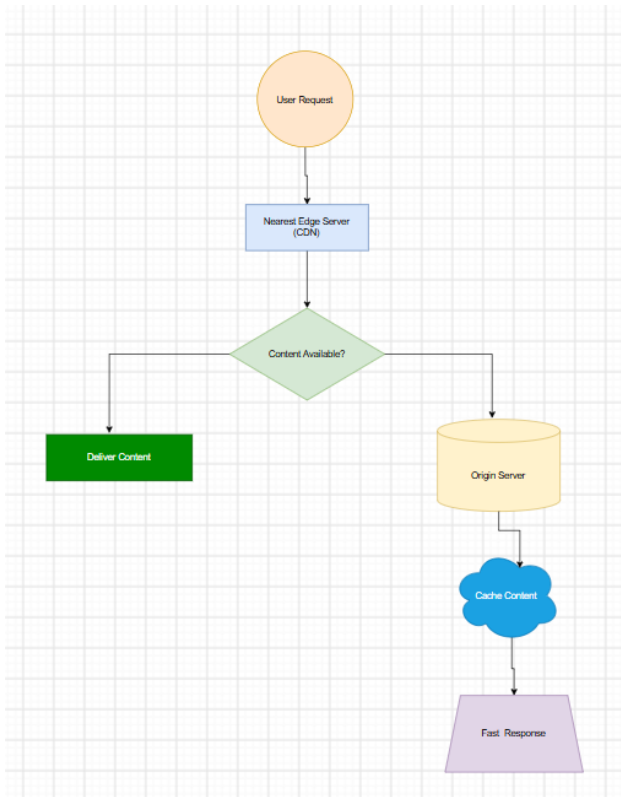
Features include:

- Multi-region deployment.
- Automatic traffic routing.
- Regional failover.

Example

Netflix stores video content on CDN edge servers worldwide. When users watch a movie, the content is delivered from the nearest server instead of the main data center. This reduces buffering, improves video quality, and provides a smooth viewing experience.

Similarly, online multiplayer games use CDNs to reduce lag and improve gameplay.



4. Cloud Security and Regulatory Compliance

Financial institutions process highly confidential customer information and must comply with government regulations. Cloud providers offer advanced security and compliance services to protect sensitive financial data.

Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI DSS
- SOC 2
- GDPR

Logging

Logging records every system activity, including:

- User login attempts.
- Data access.
- File modifications.
- Transaction history.
- Administrative actions.

Logs are useful for audits and investigations.

Monitoring

Cloud monitoring continuously checks:

- Unauthorized access attempts.
- Suspicious activities.
- Resource utilization.
- Security threats.
- System failures.

Automatic alerts help administrators respond quickly.

Encryption

Encryption protects financial information during storage and transmission.

Types include:

- AES-256 for stored data.
- TLS/SSL for transmitted data.
- Key Management Services (KMS) for encryption key control.

Data Sovereignty

Some countries require customer data to remain within national borders.

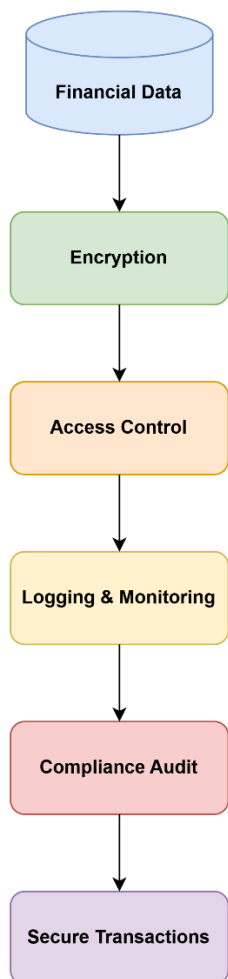
Cloud providers allow organizations to:

- Choose specific cloud regions.

- Restrict data movement.
- Meet local legal requirements.
- Maintain compliance with national regulations.

Financial Transaction Example

An online banking system stores customer account details in an Indian cloud region. Every transaction is encrypted, logged, monitored continuously, and audited regularly. Multi-factor authentication prevents unauthorized access, ensuring compliance with financial regulations and protecting customer information.



5. Containers and Kubernetes

Modern software development increasingly uses **containers** instead of traditional virtual machines because they are lightweight, portable, and efficient.

Containers vs Virtual Machines

Containers

- Share the host operating system.
- Start within seconds.
- Consume fewer resources.
- Require less storage.
- Suitable for cloud-native applications.
- Easy to deploy and update.

Virtual Machines

- Include a complete guest operating system.
- Require more memory and CPU.
- Take longer to start.
- Consume larger storage space.
- Better for complete OS isolation.

Kubernetes

Kubernetes is a container orchestration platform that automates:

- Container deployment.
- Load balancing.
- Auto-scaling.
- Service discovery.
- Health monitoring.
- Automatic recovery.

Portability

Containers package the application and all dependencies together.

Therefore, the same container can run on:

- AWS
- Microsoft Azure
- Google Cloud Platform
- Private cloud
- Hybrid cloud
- Local development machines

Microservices Architecture

Instead of building one large application, developers divide applications into small independent services.

Example of an online shopping application:

- User Authentication Service
- Product Catalog Service
- Shopping Cart Service
- Inventory Service
- Payment Service
- Order Management Service
- Notification Service

Advantages

- Faster deployment.
- Better scalability.
- Efficient resource utilization.

- High availability.
- Easy maintenance.

